# Fusing AMF and Multi-Baseline AMF-ATI Phases for SAR Ground Moving Target Detection

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***Abstract*—This paper investigates a moving target detection metric that fuses multi-baseline signal information in along-track interferometry (ATI) synthetic aperture radar(SAR) systems to improve detection performance for slowly-moving and weak targets. For a multichannel SAR system, the proposed method first estimates the full-aperture adaptive matched filter (AMF) output and multi-baseline AMF-ATI phases between clutter-suppression outputs of different baseline pairs in the range-Doppler (or SAR) images. To suppress noise contamination, a pseudo signal with power higher than the noise floor yet lower than that of the target is introduced to stabilize each AMF-ATI phase around zero radians in the absence of moving targets. Based on these results, the local detection decision can be determined on each test. Then, a global detection test is developed by optimally fusing the local decisions according to the Chair-Varshney fusion rule. Through Monte Carlo simulations, the proposed detector presents significant advantages in detecting slow and weak targets.**

***Index Terms*—Slow-moving weak target detection, Along-Track Interferometry (ATI), optimal fusion.**

## I. Introduction

Airborne or spaceborne synthetic aperture radar (SAR) ground moving target identification (GMTI) systems often deploy antenna arrays with multiple phase centers to suppress ground clutter and thereby enhance signal-to-clutter-plus-noise ratio (SCNR) for detecting moving targets within strong clutter [1]. However, moving target detection solely based on the clutter-suppressed magnitude may have a high false alarm rate in complex ground environments due to the limited independent and identically distributed (i.i.d.) clutter samples in practice [2]–[5]. Increasing the magnitude detection threshold can reduce the false alarms, while it may fail to detect the weak and slow targets [6], [7]. Up to now, it is still a hot and difficult topic to accurately detect moving targets in heterogeneous clutter backgrounds.

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The SAR along-track interferometric(ATI) interferometry measures the signal difference between dual-channel SAR images, where the ATI phase is theoretically proportional to the line of velocity of the backscatter in SAR images [8]. Based on this property, ATI technologies have been widely used to discriminate slowly-moving targets from clutter backgrounds. Recently, various two-step detectors by combining clutter suppression method with ATI techniques have been proposed [9], [10] for SAR-GMTI. Typically, the two-stage methods first construct a amplitude test based on clutter-suppression residuals to detect potential moving targets, such as adaptive matched filter (AMF) [11], and then, further use different statistics, such as ATI phase [10] and radial velocity consistency [12], to refine the detection results. Owing to exploiting both the magnitude and phase information, they can acquire an acceptable minimum detectable velocity (MDV) during moving target detection in heterogeneous terrains. In the second stage of the above-mentioned methods, the utilization of spatial degrees of freedom (DoFs) may be restricted for multichannel systems, which constrains the improvements in target detection performance. In [13], an optimal fusion-based detection method was proposed to fully exploit spatial DoFs, significantly improving MDV performance in multichannel SAR systems. However, since target signals coexist with clutter during ATI measurements, the estimation accuracy of ATI phases degrades for low-SCNR targets. Although [14] performs ATI measurements after adaptive matched filtering to suppress clutter, noise-dominated interference phases after suppression introduce strong phase randomness, which directly leads to significant detection performance degradation.

To address this issue, an improved method that fuses AMF and multi-baseline AMF-ATI phases for SAR ground moving target detection is developed. First, adaptive clutter suppression is applied to SAR (or range-Doppler) images from different baselines to suppress ground clutter and generate residual images. Based on the magnitude of the residual image processed with full spatial DoFs, the AMF test is formulated, while multi-baseline AMF-ATI phases are constructed by performing complex ATI between residual images from different baselines. During ATI processing, a pseudo signal with power higher than the noise floor but lower than that of the target is introduced, which stabilizes the multi-baseline AMF-ATI phases around zero radians in the absence of moving targets and thus enables target detection from the background. Finally, the AMF test and multi-baseline AMF-ATI phases are fused into a global test based on the Chair-Varshney criterion [15]. Simulation results demonstrate that the proposed detector outperforms state-of-the-art methods in detecting dim targets under heterogeneous clutter backgrounds.

Notations: $[\cdot]^{\mathrm{T}}$ and $[\cdot]^{\mathrm{H}}$ represent the transposition, and the complex conjugate transposition, respectively. $\arg[\cdot]$ denotes the phase of a complex number, ranging in $[-\pi, \pi]$, and $j$ is the imaginary unit with $j^2 = -1$.

## II. Signal model

Consider an airborne SAR system equipped with a linear array that consists of $M$ antenna elements uniformly distributed along the flight direction. Assume the aircraft moves at a linear velocity $v_{\mathrm{p}}$ along the track direction, the spacing between adjacent channels is $d$ and the radar wavelength is $\lambda$. The system operates in a side-looking mode. After signal processing, which includes matched filtering, coherent integration, image registration, and channel calibration, the constant amplitude and phase differences between SAR images caused by channel response errors are eliminated [7]. The data for a given pixel $q$ from the $M$ SAR images is represented by an $M \times 1$ vector $\mathbf{z}(q) = [z_1(q), \ldots, z_M(q)]^{\mathrm{T}}$. Accordingly, the binary hypothesis test for target detection is defined as follows:

$$\begin{aligned} &\mathrm{H}_0 : \mathbf{z}(q) = \mathbf{c}(q) + \mathbf{n}(q), \\ &\mathrm{H}_1 : \mathbf{z}(q) = \mathbf{s}(q) + \mathbf{c}(q) + \mathbf{n}(q), \end{aligned} \tag{1}$$

where $\mathrm{H}_0$ is the target-absent hypothesis and $\mathrm{H}_1$ is the target present hypothesis, $\mathbf{s}(q)$, $\mathbf{c}(q)$, and $\mathbf{n}(q)$ are the target signal, clutter, and noise vectors, respectively.

Under hypothesis $H_1$, the target signal vector $\mathbf{s}(q)$ is expressed as:

$$\mathbf{s}(q) = \beta_{\mathrm{t}}(q)\boldsymbol{\alpha}_{\mathrm{t}}(q), \tag{2}$$

Here, $\beta_{\mathrm{t}}(q)$ is the target complex amplitude and $\boldsymbol{\alpha}_{\mathrm{t}}(q)$ is the target spatial steering vector, which is expressed as:

$$\boldsymbol{\alpha}_{\mathrm{t}}(q) = \left[1, \ldots, \exp\left(j2\pi f_{\mathrm{t}}(q)\frac{(M-1)d}{2v_{\mathrm{p}}}\right)\right]^{\mathrm{T}}, \tag{3}$$

where $f_{\mathrm{t}}(q) = 2v_{\mathrm{d}}(q)/\lambda$ is the target's Doppler frequency determined by its radial velocity $v_{\mathrm{d}}(q)$.

For heterogeneous environments, the clutter is described by the product model [9], [10], [12], [16]:

$$\mathbf{c}(q) = \xi(q) \times \gamma(q)\boldsymbol{\alpha}_{\mathrm{c}}(q), \tag{4}$$

where $\gamma(q)$ denotes the complex Gaussian speckle and $\xi(q)$ represents the texture variable to describe the backscatter variations [10], [12], [14]. For stationary ground clutter, $\boldsymbol{\alpha}_{\mathrm{c}}(q) \approx [1, \ldots, 1]^T$.

## III. Global optimal fusion detection

In this section, we present a detector fusing AMF and multi-baseline AMF-ATI phases for SAR GMTI. The functional block diagram of this method is shown in Fig. 1. In this method, the AMF test ($T$) is based on the clutter-suppression residuals with the $M$-channel SAR images, meanwhile, the multi-baseline AMF-ATI phase $(\varphi_1, \varphi_2, \cdots, \varphi_{M-2})$ are measured by applying the ATI processing between two pseudo-signal-aided residual images from clutter suppression with different baselines. Given the local detection thresholds $\eta_0, \eta_1, \cdots, \eta_{M-2}$, the local decisions $u_0, u_1, \cdots, u_{M-2}$ are then decided. These decisions are subsequently fused by the optimal weights $a_0, a_1, \cdots, a_{M-2}$ to form a global test. The detailed formulation of the proposed method is presented in the remainder of this section.

### A. Local Detection

*1) AMF Detection:* Under the assumption of heterogeneous clutter and a Gaussian noise background, the array output is optimally combined in the SAR image via the AMF framework using an $M \times 1$ optimal weight vector $\mathbf{u}(q)$. The clutter suppression output is $y(q) = \mathbf{u}^{\mathrm{H}}(q)\mathbf{z}(q)$, where the optimal weight vector is given by $\mathbf{u}(q) = (\mathbf{R}^{-1}(q)\boldsymbol{\alpha}_{\mathrm{t}}(q))/(\boldsymbol{\alpha}_{\mathrm{t}}^{\mathrm{H}}(q)\mathbf{R}^{-1}(q)\boldsymbol{\alpha}_{\mathrm{t}}(q))$, The CCM $\mathbf{R}(q)$ is estimated from $K$ surrounding i.i.d. training samples as $\hat{\mathbf{R}}(q) = (1/K)\sum_{k=1}^{K}\tilde{\mathbf{z}}_q(k)\tilde{\mathbf{z}}_q^{\mathrm{H}}(k)$, where $\tilde{\mathbf{z}}_q(k) = [z_1(k), z_2(k), \ldots, z_M(k)]^{\mathrm{T}}$ denotes the snapshot of pixel $q$ with the i.i.d. sample $k$. The amplitude test based on the residual amplitude is then constructed as

$$T = \frac{\mathbf{u}^{\mathrm{H}}(q)\mathbf{z}(q)\mathbf{z}^{\mathrm{H}}(q)\mathbf{u}(q)}{\sigma_{\mathrm{cn}}} \overset{H_1}{\geq} \eta_0, \tag{5}$$

where $\sigma_{\mathrm{cn}}$ denotes the average power after adaptive clutter suppression, and $\eta_0$ denotes the local threshold. For this local test, hypothesis $\mathrm{H}_1$ is accepted if the test statistic $T$ is greater than or equal to the threshold; otherwise, hypothesis $\mathrm{H}_0$ is accepted.

*2) Multi-baseline AMF-ATI Detection:* Adaptive match filter is first applied on SAR (or range-Doppler) images from different baselines to suppress ground clutter, outputting corresponding residual images. For example by using SAR images from the first and last $M-1$ channels, the $(M-1)\times 1$ filter $\mathbf{u}_1(q)$ is applied on two $(M-1)\times 1$ data vectors:

$$y_1(q) = \mathbf{u}_1^{\mathrm{H}}(q)\mathbf{z}_1(q), \tag{6a}$$

$$y_2(q) = \mathbf{u}_1^{\mathrm{H}}(q)\mathbf{z}_2(q), \tag{6b}$$

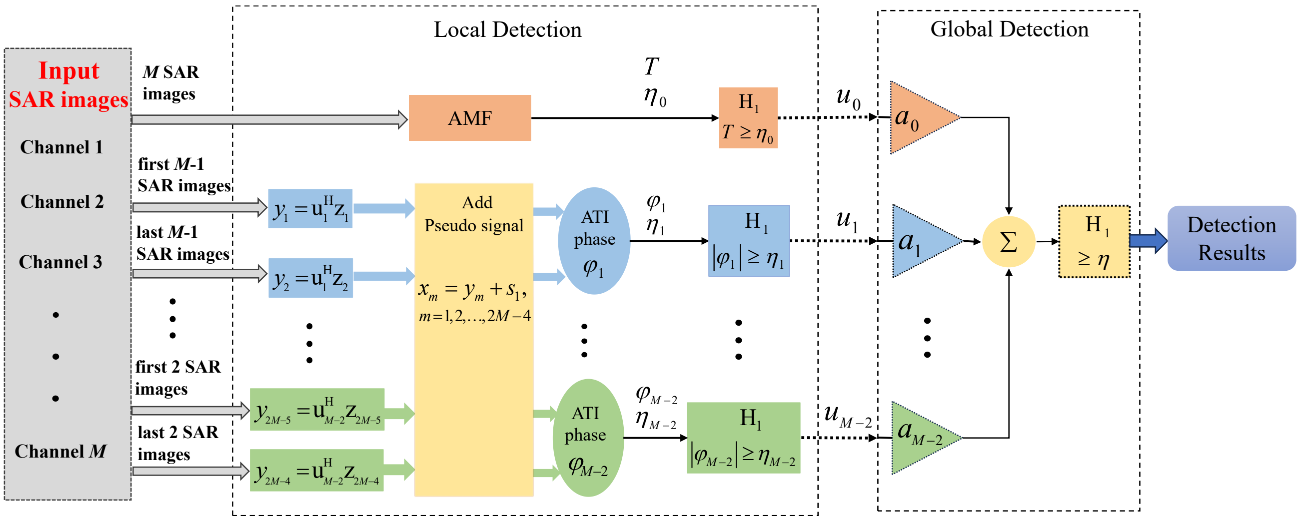



Fig. 1: Functional block diagram of the proposed detection method.

where $\mathbf{z}_1(q) = [z_1(q), \ldots, z_{M-1}(q)]^{\mathrm{T}}$, $\mathbf{z}_2(q) = [z_2(q), \ldots, z_M(q)]^{\mathrm{T}}$, and $\mathbf{u}_1(q)$ is the optimal weight, as given by

$$\mathbf{u}_1(q) = \frac{\mathbf{R}_1^{-1}(q)\boldsymbol{\alpha}_{\mathrm{t}1}(q)}{\boldsymbol{\alpha}_{\mathrm{t}1}^{\mathrm{H}}(q)\mathbf{R}_1^{-1}(q)\boldsymbol{\alpha}_{\mathrm{t}1}(q)}, \tag{7a}$$

$$\hat{\mathbf{R}}_1(q) = \frac{1}{K}\sum_{k=1}^{K}\tilde{\mathbf{z}}_{1,q}(k)\tilde{\mathbf{z}}_{1,q}^{\mathrm{H}}(k), \tag{7b}$$

$$\boldsymbol{\alpha}_{\mathrm{t}1}(q) = \left[1, \ldots, \exp\left(j2\pi f_{\mathrm{t}}(q)\frac{(M-2)d}{2v_{\mathrm{p}}}\right)\right]^{\mathrm{T}}, \tag{7c}$$

where $\hat{\mathbf{R}}_1(q)$ is the estimation of CCM with sample data from the first $M-1$ channels and $\tilde{\mathbf{z}}_{1,q}(k) = [z_1(k), z_2(k), \ldots, z_{M-1}(k)]^{\mathrm{T}}$ denotes the snapshot of pixel $q$ with the i.i.d. sample $k$, being the $(M-1)\times 1$ data vector, while $\mathbf{a}_{\mathrm{t}1}(q)$ is the target spatial steering vector across the $M-1$ channels.

By introducing a pseudo signal $s_1$ with power higher than the noise floor yet lower than that of the target in two residual images, such as $y_1$ and $y_2$, the AMF-ATI phase $\varphi_1$ is formulated as

$$\varphi_1(q) = \arg[(y_1(q) + s_1)(y_2^*(q) + s_1)]. \tag{8}$$

Assume that clutter is effectively suppressed in adaptive matched filter, and thus, moving targets with magnitude larger than that of noise and pseudo signal can play a dominant role in the interferometry operation, leading to

$$\varphi_1(q) \approx 2\pi f_{\mathrm{t}}(q)\frac{d}{2v_{\mathrm{p}}}, \tag{9}$$

which is approximately proportional to the target Doppler frequency $f_{\mathrm{t}}(q)$. While, under $\mathrm{H}_0$, when clutter residuals present due to heterogeneous clutter, the corresponding interferometric phase $\varphi_1(q)$ would be approximately 0 as the Doppler frequency is zero. In another case of hypothesis $\mathrm{H}_0$, clutter residuals may be close to the noise signal for homogeneous clutter. As such, the pseudo signal can play a dominant role in the interferometry operation, leading to the interferometric phase of non-target around zero rad.

Similarly, by Using (8) on any other two clutter-suppression outputs from different baselines, a set of AMF-ATI phases can be obtained as $\varphi_1(q), \varphi_2(q), \cdots, \varphi_{M-2}(q)$. Based on these multi-baseline AMF-ATI phases, the local detection is formulated as

$$\varphi_n(q) \overset{\mathrm{H}_1}{\geq} \eta_n, n = 1, 2, \cdots, M-2, \tag{10}$$

where $\eta_n$ denotes the detection threshold corresponding to the $n$-th AMF-ATI phase test, and the hypothesis $\mathrm{H}_1$ is accepted when $\varphi_n(q)$ exceeds or is equal to $\eta_n$ during this local detection. Otherwise, hypothesis $\mathrm{H}_0$ is accepted.

### B. Global Detection

For each pixel under test, the above-mentioned local detectors can yield $M-1$ binary decisions $u_0, u_1, \cdots, u_{M-2}$, where $u_i = +1$ if hypothesis $\mathrm{H}_1$ is declared, and otherwise $u_i = -1$, $0 \leq i \leq M-2$. The global test is formulated as

$$\sum_{i=0}^{M-2} a_i u_i \overset{\mathrm{H}_1}{\geq} \eta, \tag{11}$$

where $\eta$ means global detection threshold, each weight $a_i$ corresponds to each local detector's detection probability $\mathrm{P}_{\mathrm{d}i}$ and false alarm probability $\mathrm{P}_{\mathrm{f}i}$, which can be expressed as

$$a_i = \begin{cases} \log\frac{\mathrm{P}_{\mathrm{d}i}}{\mathrm{P}_{\mathrm{f}i}}, & \text{if } u_i = +1, \\ \log\frac{1-\mathrm{P}_{\mathrm{f}i}}{1-\mathrm{P}_{\mathrm{d}i}}, & \text{if } u_i = -1. \end{cases} \tag{12}$$

A target is declared present (Hypothesis H1 is accepted) if the weighted sum in (11) exceeds or is equal to $\eta$, otherwise, the target is declared absent (Hypothesis H0 is accepted).

## IV. Simulation Experiments

Simulation data were generated based on the multichannel signal model described in Section II, with radar system parameters listed in Table I. heterogeneous clutter backgrounds were constructed, comprising 700 strong clutter samples with CNR fluctuating between 10 dB and 55 dB, and 300 homogeneous background samples at a fixed CNR of 10 dB. A moving target was superimposed at the 800-th sample. To account for channel inconsistencies, random phase errors (zero mean, standard deviation of $1°$) were introduced across the channels. Additionally, the correlation coefficient $\rho$ between adjacent channels was set to approximately 0.995. Finally, this 6-channel system consists of one AMF test and four AMF-ATI phase tests.

TABLE I
RADAR PARAMETERS

| **Parameter** | Radar Wavelength ($\lambda$) | Aircraft Velocity ($v_{\mathrm{p}}$) | Channel Number ($M$) | Channel Spacing ($d$) |
|---|---|---|---|---|
| **Value** | 0.019 m | 101 m/s | 6 | 0.167 m |

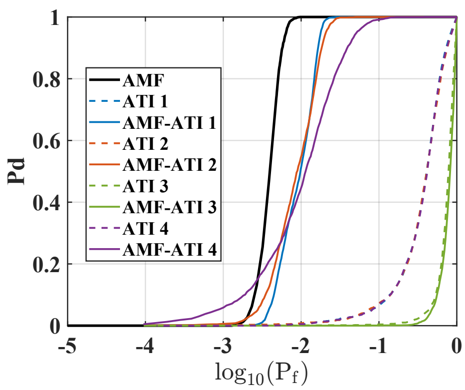
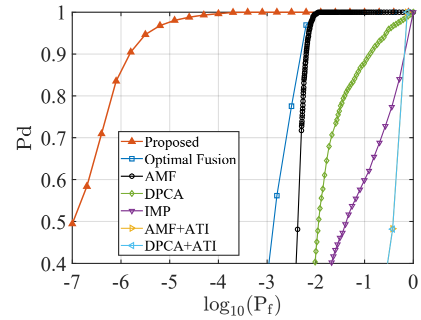


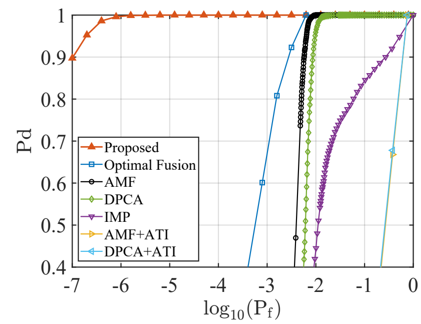

Fig. 2: $\mathrm{P_d}$ versus false alarm probability $\mathrm{P_f}$ of ATI and AMF-ATI phase tests under input SNR of 12 dB, pseudo signal's SNR of 2 dB and $v_{\mathrm{d}} = 4$ m/s.

To begin with, the effectiveness of the multi-baseline AMF-ATI phase test in Section III is verified through the Receiver Operating Characteristic (ROC) curves shown in Fig. 2. The ATI 1 through ATI 4 represent the tests based on firsthand measured interferometric phases between the first $m$ and the last $m$ SAR images, and the AMF-ATI 1 through AMF-ATI 4 represent the phase tests between the first $m$ and the last $m$ SAR images based on the method in Section III, where $m = 5, 4, 3,$ and 2, respectively. For a target with input SNR of 12 dB and radial velocity of 4 m/s, given the pseudo signal's SNR of 2 dB, the comparisons of ATI and AMF-ATI indicate that the AMF-ATI detectors achieve a significant improvement in detection performance, which means that AMF-ATI not only improves the accuracy of the estimated target interference phase but also suppresses the interference of noise on the phase.The standard ATI tests fail to adequately suppress clutter, leading to poor target detection performance under low Pfas. The ATI 3 and AMF-ATI 3 phase test, however, exhibit poor performance as the target's velocity corresponds to the system's blind velocity for that specific baseline.

Furthermore, we compare the proposed method with several state-of-the art techniques, including: the amplitude test based on AMF from [2], the displaced phase center antenna (DPCA)




(a)




(b)

Fig. 3: $\mathrm{P_d}$ versus $\mathrm{P_f}$ for the single test under input SNR of 12dB and pseudo signal's SNR of 2 dB (a) $\mathrm{P_d}$ versus $\mathrm{P_f}$ with $v_{\mathrm{d}} = 2$ m/s, (b) $\mathrm{P_d}$ versus $\mathrm{P_f}$ with $v_{\mathrm{d}} = 4$ m/s.

test from [17], the joint metric of interferometric magnitude and phase (IMP) from [10], the two-step methods combining the AMF and ATI (AMF+ATI) and the DPCA and ATI (DPCA+ATI) from [17], and the optimal fusion-based method in [13]. under setting a set of local detectors' configurations for the proposed method and the optimal fusion method and a series of thresholds for other methods, Figs. 3(a) and Figs. 3(b) present the ROC comparison depicting Pd versus Pfa between the proposed method and other methods with $v_{\mathrm{d}} = 2$ m/s and $v_{\mathrm{d}} = 4$ m/s, respectively. The proposed method consistently demonstrates superior target detection performance compared to other methods. Then followed by the optimal fusion method, though it fully utilize the DoFs of this 6-channel system, the optimal fusion method is ultimately capped by the poor reliability of its ATI phase tests. The poor performance of the two-step methods are also affected by the ATI test in the second step, which is severely disrupted by the strong clutter and noise.

## V. Conclusion

In this work, a novel target detector that optimally fuses AMF and multi-baseline AMF-ATI phases for SAR GMTI in heterogeneous clutter backgrounds is developed. As the phase estimation accuracy for low-SCNR targets is enhanced in the AMF-ATI phase, and full spatial DoFs are exploited in the detection, the proposed method achieves significant improvements in the detection of dim targets in heterogeneous clutter backgrounds. Preliminary simulation results demonstrate that the proposed detector can exhibit superior detection capability compared with existing methods.

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